

Project Design Phase-II Data Flow Diagram & User Stories

Date	01 October 2026
Team ID	SWTID-2026-8331
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

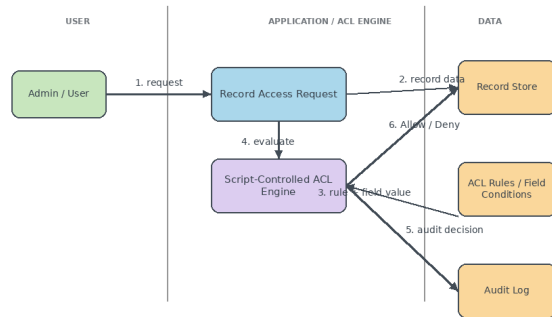
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a visual representation of how information moves through the Script-Controlled ACL system. It shows how a record-access request enters the system, how the field value and ACL rule are evaluated, where rules and audit results are stored, and how an Allow or Deny decision is returned.

Example: Script-Controlled ACL – Simplified DFD

Example: DFD Level 0 (Industry Standard)

Flow



1. User requests a record operation.
2. Application loads the target record and user context.
3. ACL rule reads the configured field value and condition.
4. Script-Controlled ACL evaluates the condition.
5. Allow / Deny decision and reason are recorded.
6. The permitted operation proceeds; denied operations are blocked.

User Stories

Use the below template to list the user stories for Script-Controlled ACL.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Administrator	ACL Rule Management	USN-1	As an administrator, I can create an ACL rule that checks a record field value.	The rule can be saved with field, condition, and action details.	High	Sprint-1
Administrator	ACL Rule Management	USN-2	As an administrator, I can update an existing field-value ACL rule.	Updated rule is stored and applied to subsequent requests.	High	Sprint-1
Administrator	ACL Rule Management	USN-3	As an administrator, I can disable an ACL rule when it is no longer required.	Disabled rule is not evaluated for new access requests.	Medium	Sprint-2
Authorized User	Record Access	USN-4	As an authorized user, I can request access to a protected record.	The request reaches the ACL evaluation process with user and record context.	High	Sprint-1
Authorized User	Record Access	USN-5	As an authorized user, I can read a record when its field value satisfies the ACL condition.	The system returns the permitted record data.	High	Sprint-1
Authorized User	Record Access	USN-6	As an authorized user, I cannot update a record when its field value fails the ACL condition.	The update is blocked and an access-denied result is returned.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
System	ACL Evaluation	USN-7	As the system, I can read the configured field value before deciding access.	The correct field value is passed to the rule evaluator.	High	Sprint-1
System	ACL Evaluation	USN-8	As the system, I can evaluate the field value against the configured script condition.	The evaluator produces a clear Allow or Deny decision.	High	Sprint-1
System	Audit	USN-9	As the system, I can record the ACL decision and reason.	The audit entry contains request context, decision, and evaluation reason.	Medium	Sprint-2
Administrator	Audit	USN-10	As an administrator, I can review ACL evaluation results for a record request.	Relevant audit entries can be viewed for troubleshooting and review.	Medium	Sprint-2
Authorized User	Error Handling	USN-11	As an authorized user, I can see a clear message when access is denied.	The response explains that the record does not satisfy the access condition without exposing sensitive rule details.	Medium	Sprint-2
Administrator	Security	USN-12	As an administrator, I can test an ACL rule before enabling it.	A test result shows the expected Allow or Deny outcome for sample field values.	Low	Sprint-3